

EXPERIMENT (6)

DOUBLE IMMUNODIFFUSION AND SINGLE RADIAL IMMUNODIFFUSION TECHNIQUE (MANCINI TEST)

**IMMUNOLOGY & SEROLOGY LABORATORY
(BT336)**

PRECIPITATION REACTION

- **Precipitation reactions are based on the interaction of antibodies and antigens.**
- **They are based on two soluble reactants that come together to make one insoluble product, the precipitate.**
- **These reactions depend on the formation of (cross-links) when antigen and antibody exist in optimal proportions.**

Antigen (soluble) + Antibody (soluble) → Ag-Ab complex (insoluble)

- **Excess of either component (Ag or Ab) reduces cross-link formation and subsequent precipitation.**
- **Precipitation reactions can be performed in semisolid media such as agar or agarose, or non-gel support media such as cellulose acetate.**

PRECIPITATION REACTION

Precipitation reactions differ from agglutination reactions in the

1. Size

- **Antigens are larger in size in precipitation reactions**
- **Antigens are smaller in size in agglutination reactions**

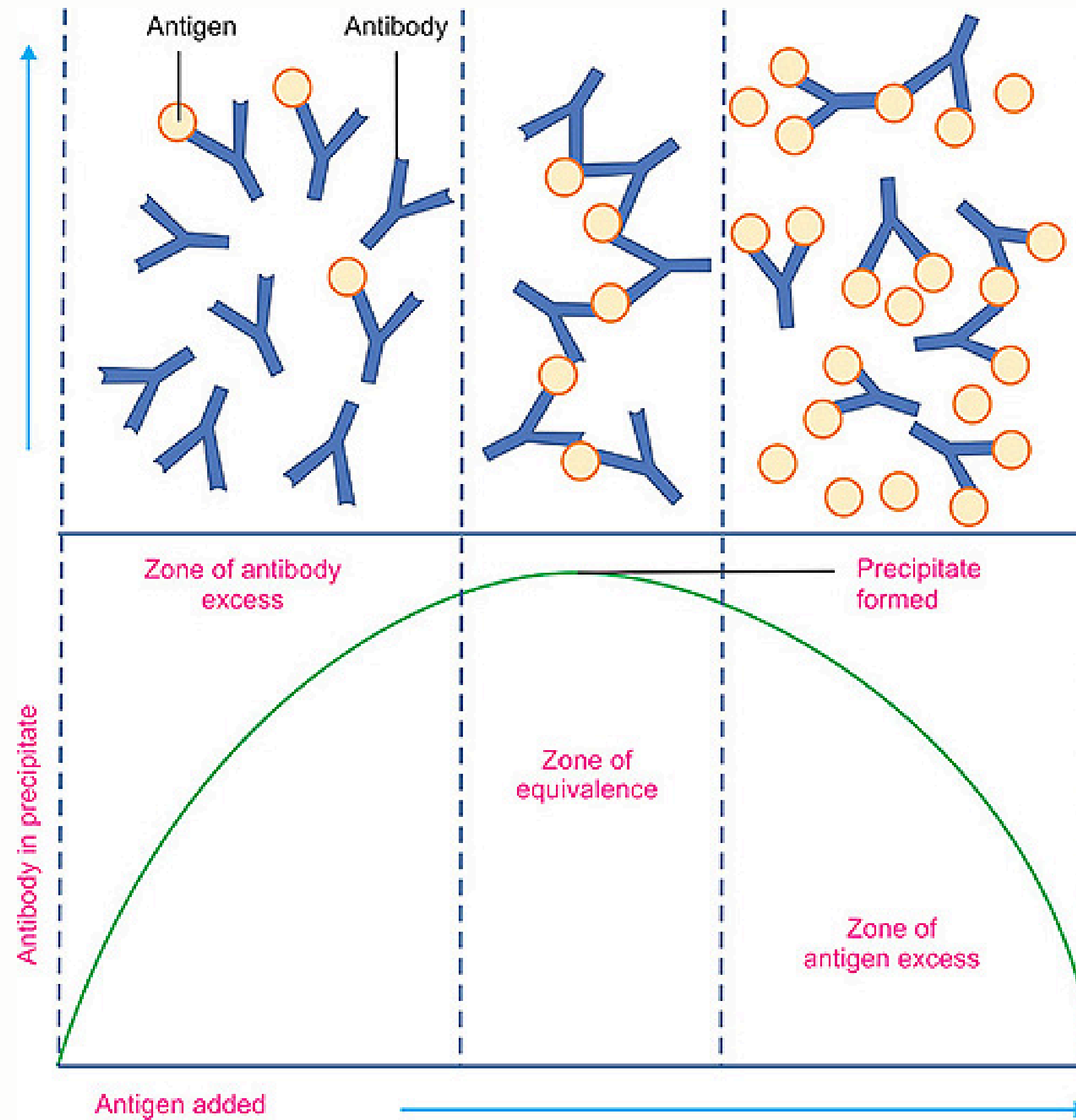
2. Solubility of the antigen

- **Antigens are soluble molecules in the precipitation reactions**
- **Antigens are insoluble (particulate) molecules in the agglutination reactions**

3. Sensitivity

- **Agglutination reaction is comparatively more sensitive than the precipitation reaction**

PRECIPITATION REACTION CURVE

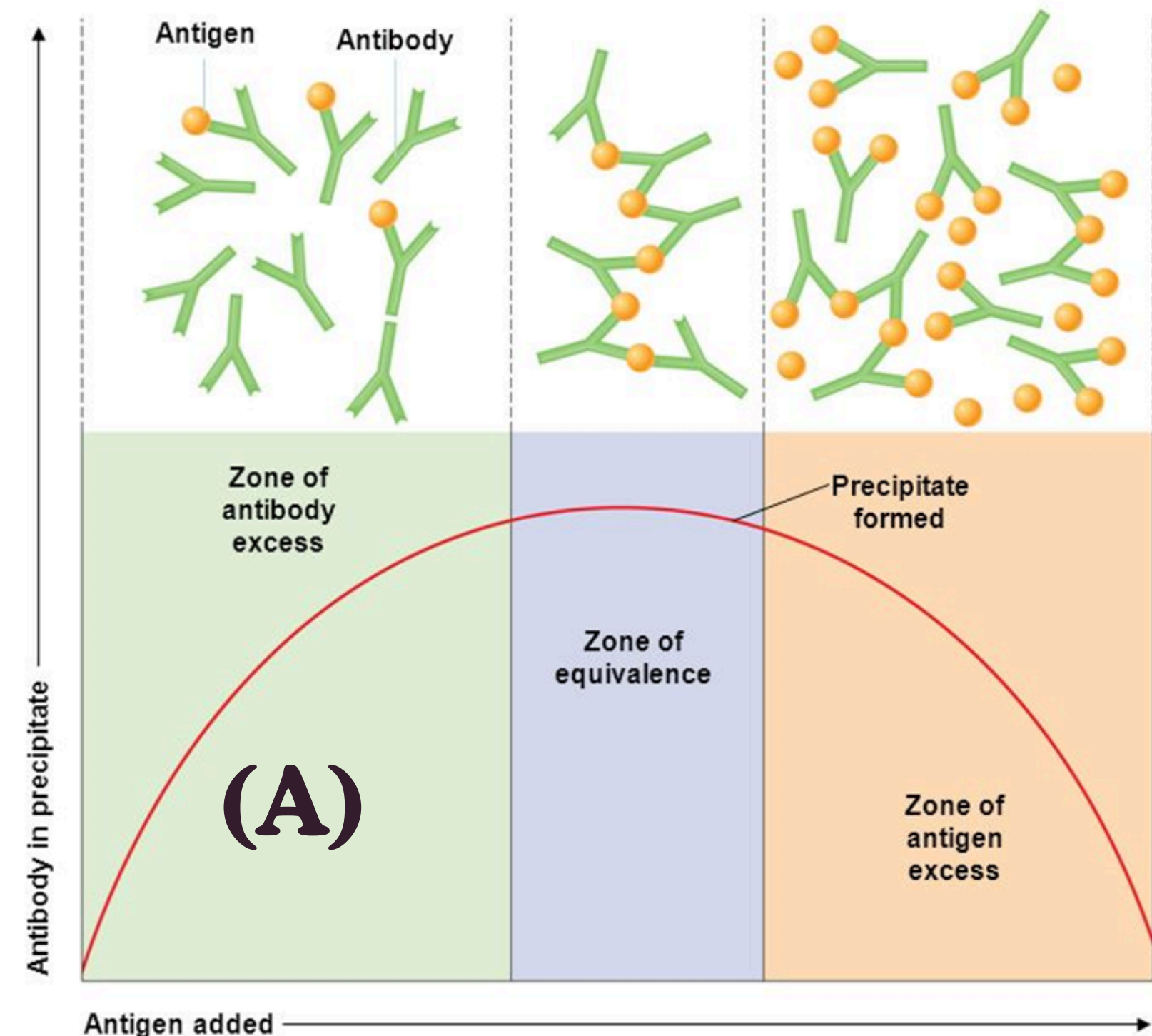


(A) PROZONE

- In this region, the concentration of antibodies is in excess compared to the antigen. As a result, not all antigens can find available binding sites on the antibodies, leading to reduced precipitation

[Ag] >> [Ab] (Ag is lower than Ab)

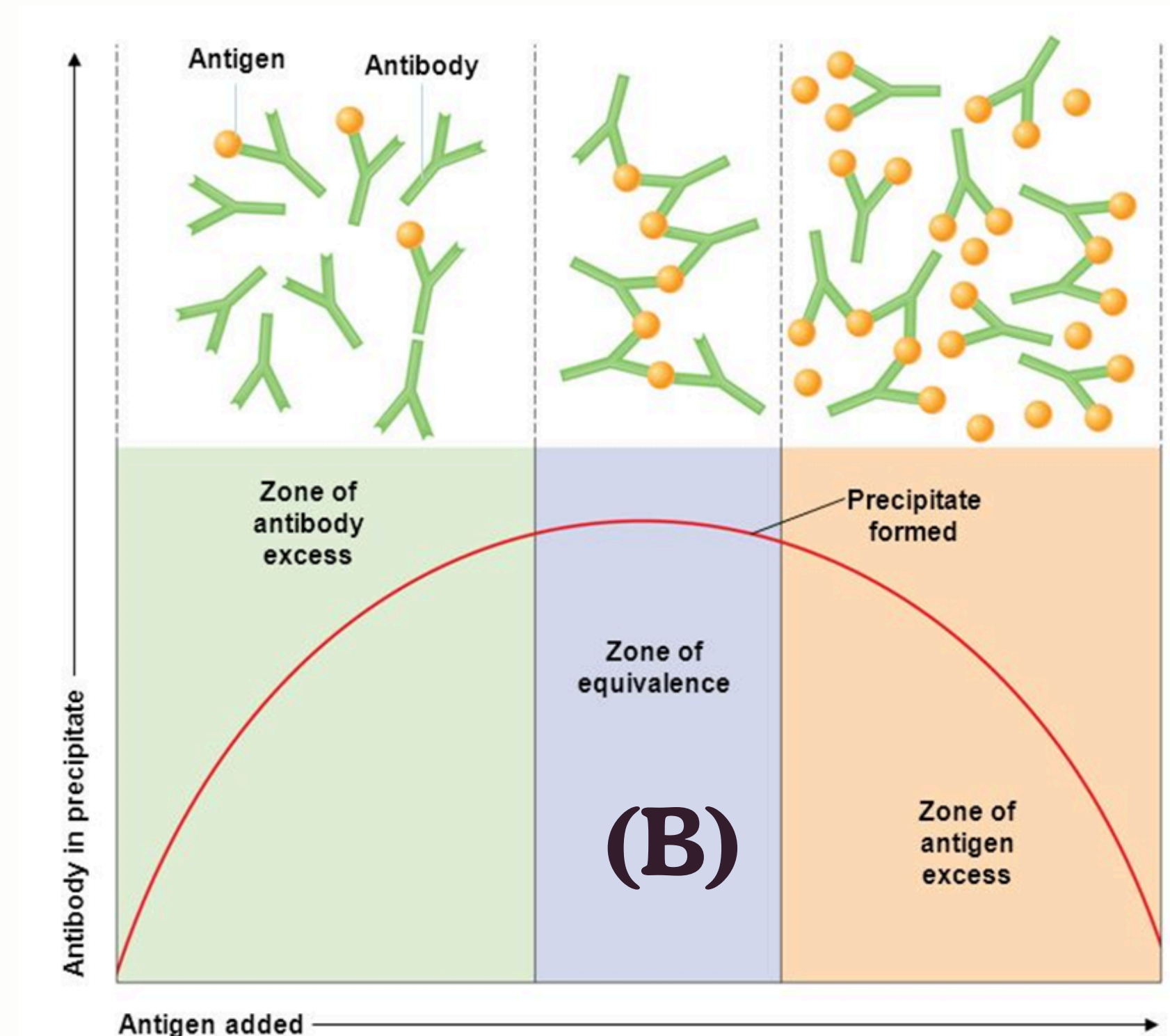
- There are cross-linkages between Ag and Ab but at very low levels, so the precipitate reaction is not visible



(B) ZONE OF EQUIVALENCE

- This is the region where antigen and antibody are present in roughly equal concentrations, leading to optimal complex formation and maximum precipitation. This zone represents the ideal conditions for the reaction.

$$[Ag] \approx [Ab]$$

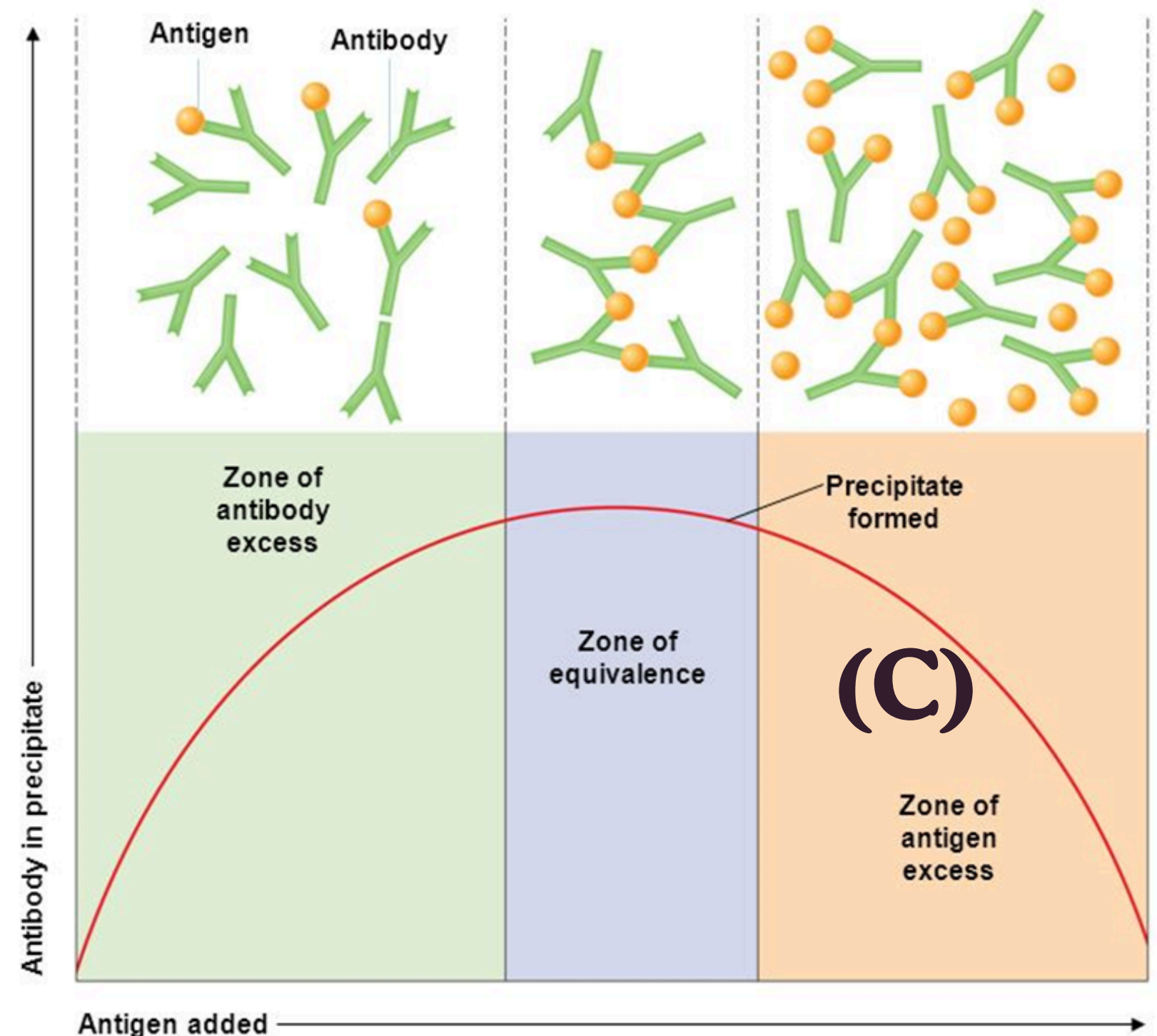


(C) POSTZONE

- In the postzone, the concentration of antigen is in excess compared to the antibody. This leads to fewer available binding sites on the antibodies for antigen binding, also resulting in reduced precipitation

$[Ag] \ll [Ab]$ (Ag is higher than Ab)

- There are cross-linkages between Ag and Ab but at a very low level



PRECIPITATION REACTION

A number of currently used techniques make use of this principle

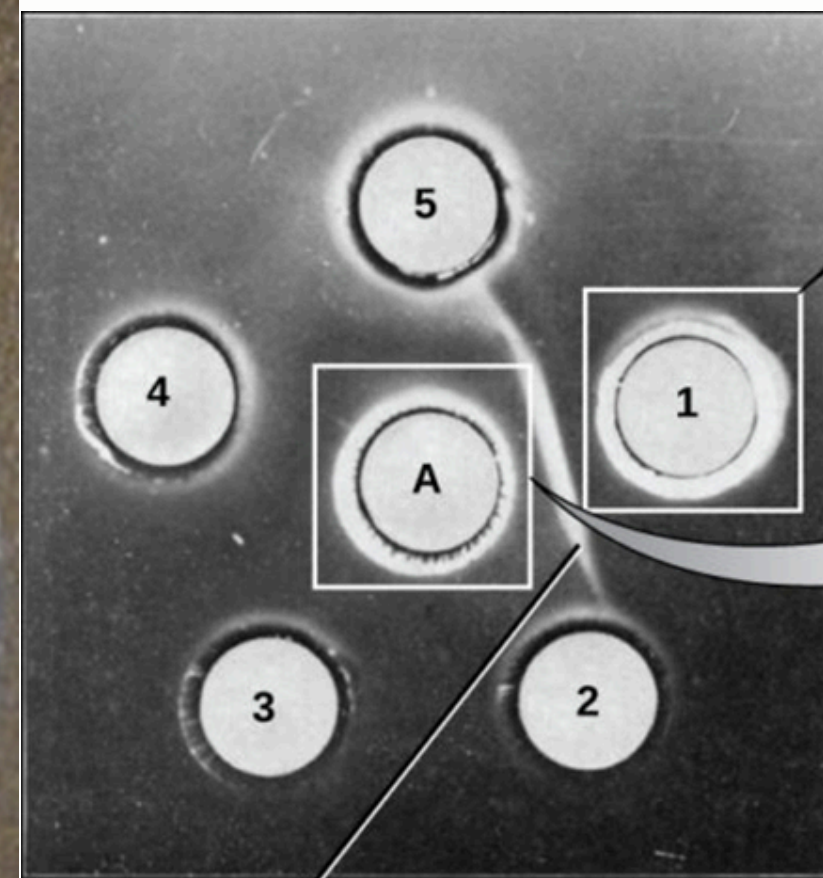
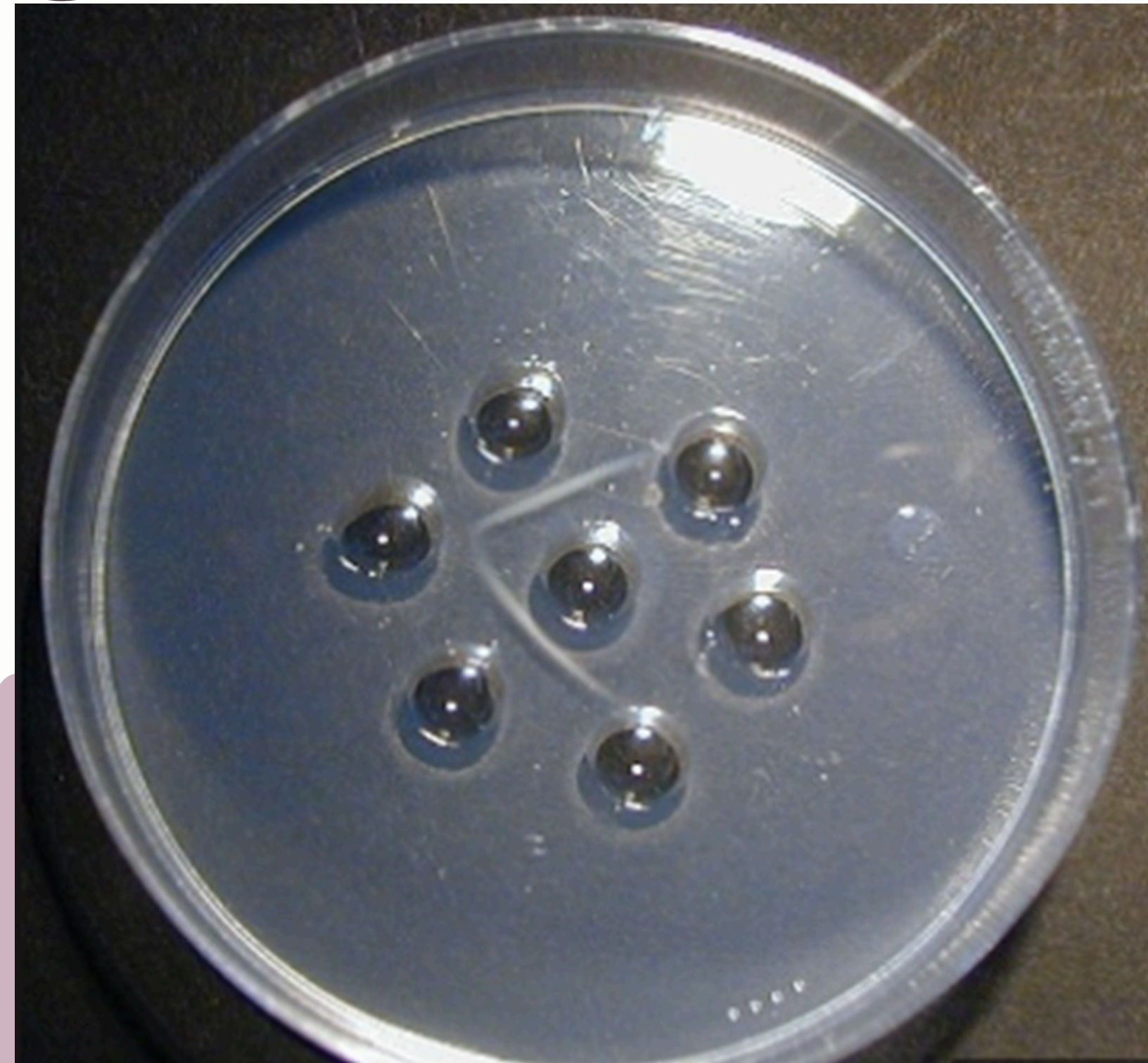
- 1. Double immunodiffusion technique**
- 2. Single radial immunodiffusion technique**
- 3. Immuno-electrophoresis technique**

DOUBLE IMMUNODIFFUSION (D.I.D) (THE OUCHTERLONY TECHNIQUE)

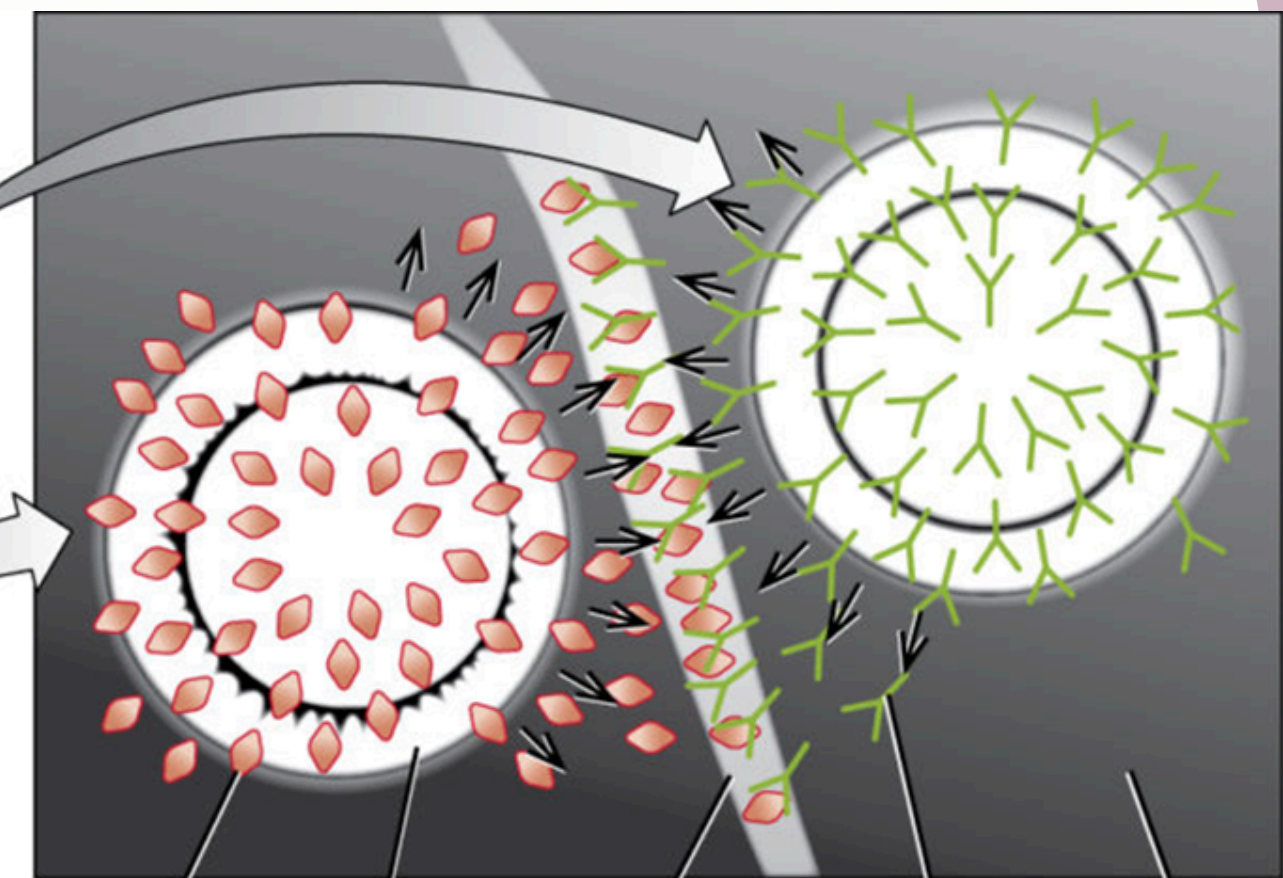
THE DOUBLE IMMUNODIFFUSION

- **Double immunodiffusion, also known as Ouchterlony double immunodiffusion, is a technique used in immunology to detect the presence of antigens or antibodies and to determine their relative concentrations.**
- **It is based on the principle that when antigen and antibody are allowed to diffuse towards each other in a gel medium, they form visible lines or bands of precipitation where they meet.**

THE DOUBLE IMMUNODIFFUSION



precipitin arc



antigen

well

precipitin arc

antibody

agar

THE DOUBLE IMMUNODIFFUSION

Double immunodiffusion is a versatile technique that can be used for various applications, including:

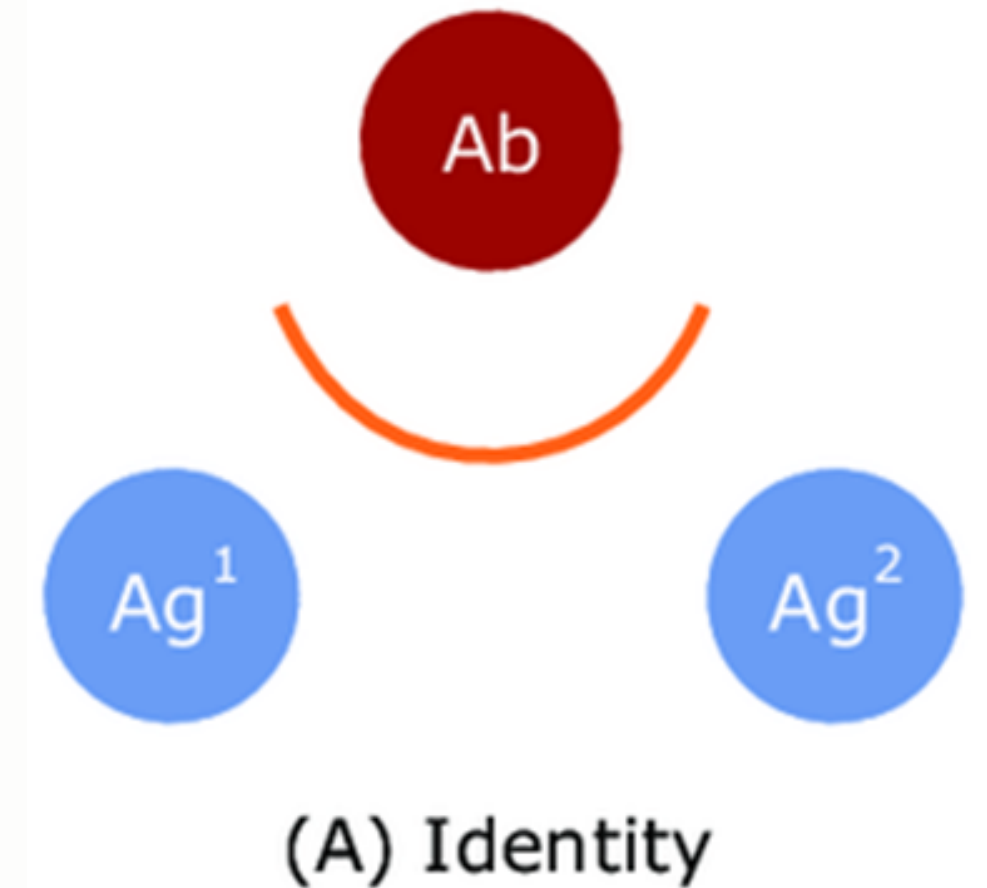
- 1. Qualitative Analysis:** It can identify the presence or absence of specific antigens or antibodies in a sample.
- 2. Quantitative Analysis:** By comparing the size and intensity of precipitation lines to standards of known concentration, the relative concentration of antigens or antibodies in a sample can be determined.
- 3. Antigen-Antibody Interaction Studies:** It can provide information about the specificity and affinity of antigen-antibody interactions.

REACTIONS OF IDENTITY

There are three basic patterns of precipitation:

(1) REACTION OF IDENTITY:

- The precipitation lines have fused completely showing the presence of identical determinants in antigens 1 and 2.

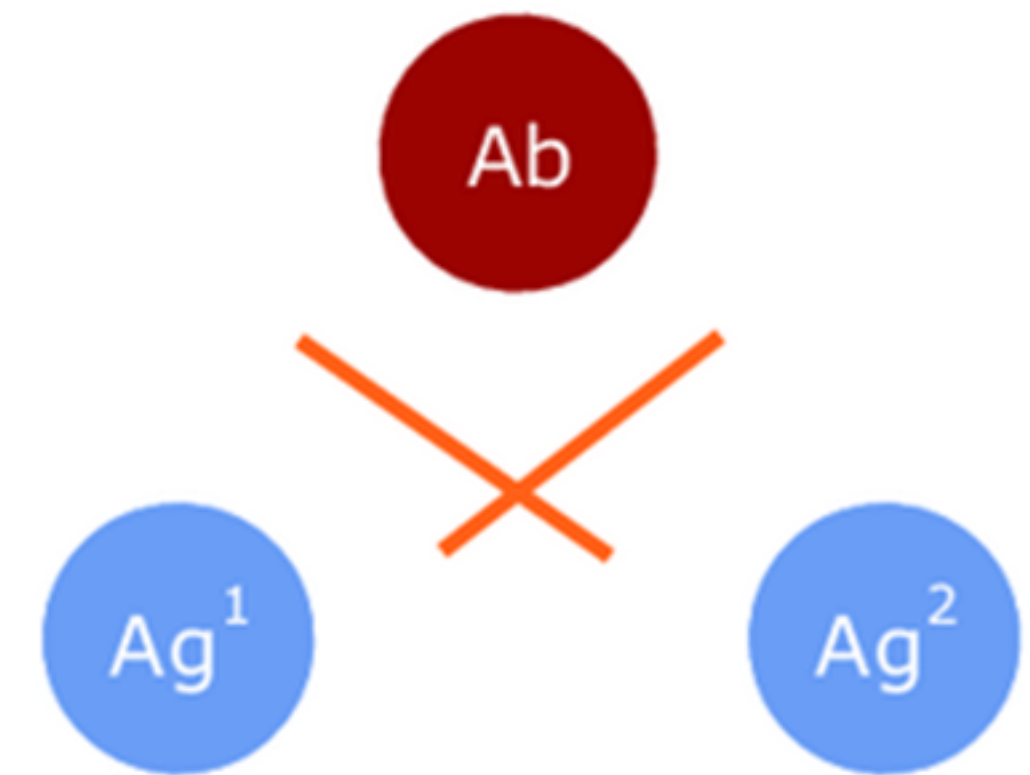


REACTIONS OF IDENTITY

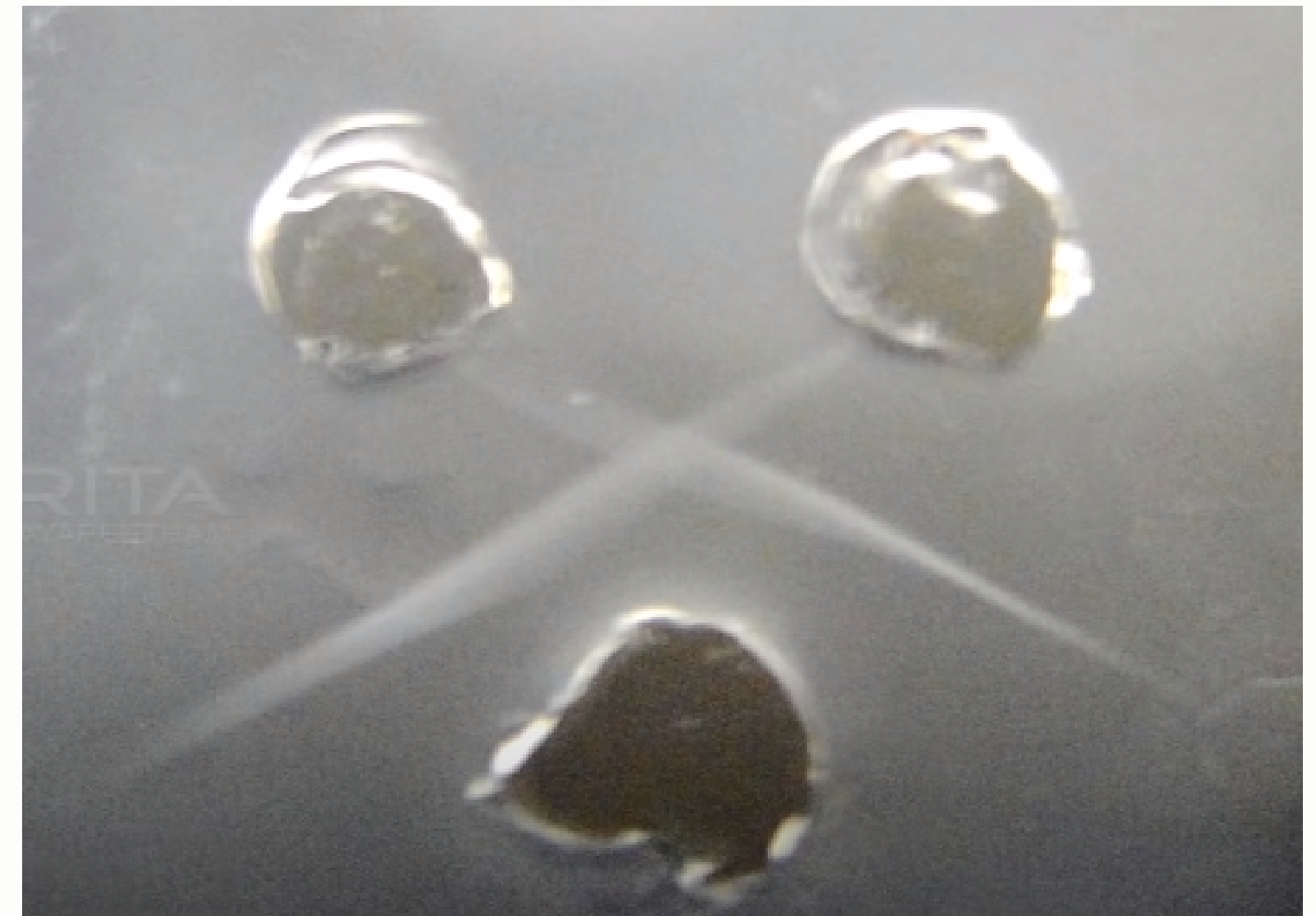
There are three basic patterns of precipitation:

(2) REACTION OF NON-IDENTITY:

- The two precipitation lines cross without any interaction. Antigens 1 and 2 do not share any determinants detectable.



(B) Non-identity

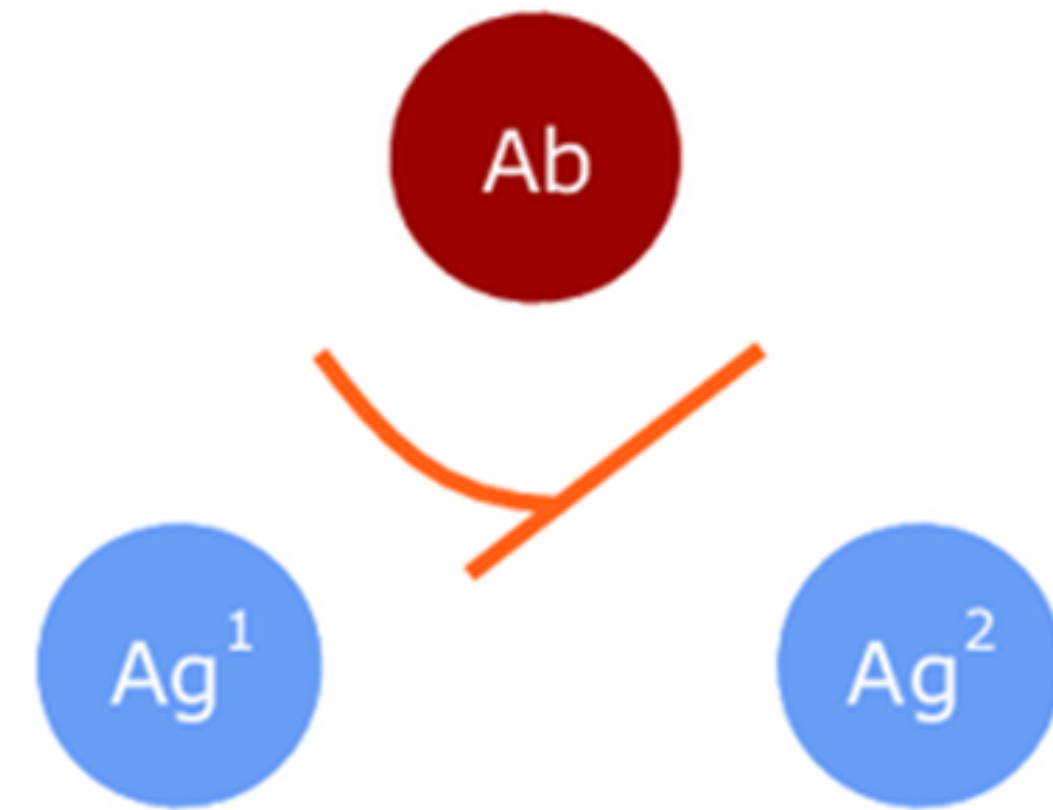


REACTIONS OF IDENTITY

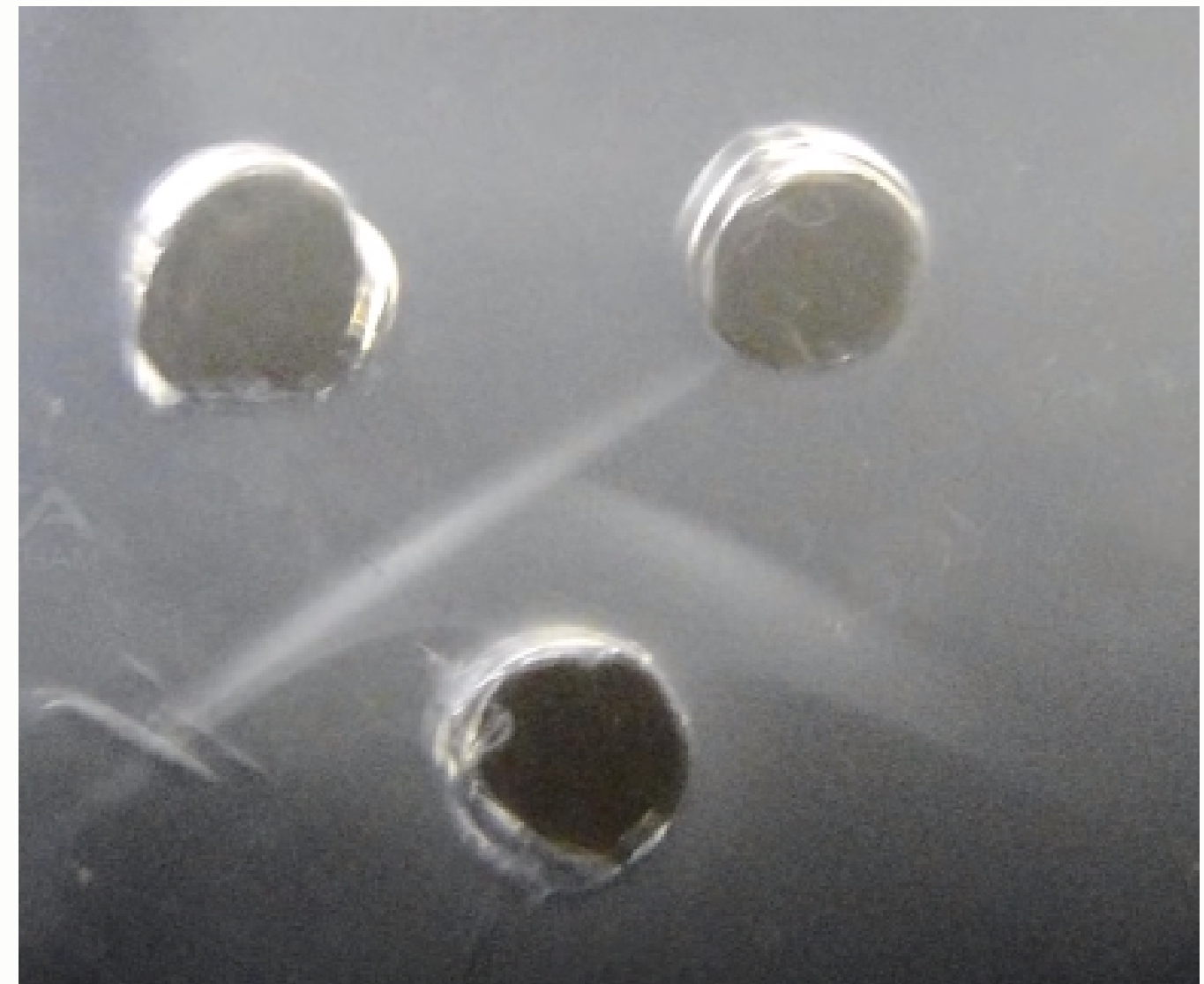
There are three basic patterns of precipitation:

(3) REACTION OF PARTIAL IDENTITY:

- The two precipitation lines have fused, but in addition, a spur has formed towards the well-containing protein 2.
- Therefore, proteins 1 and 2 share some common determinants but protein 2 has additional unique determinants



(C) Partial identity



SEMIQUANTITATIVE TESTS

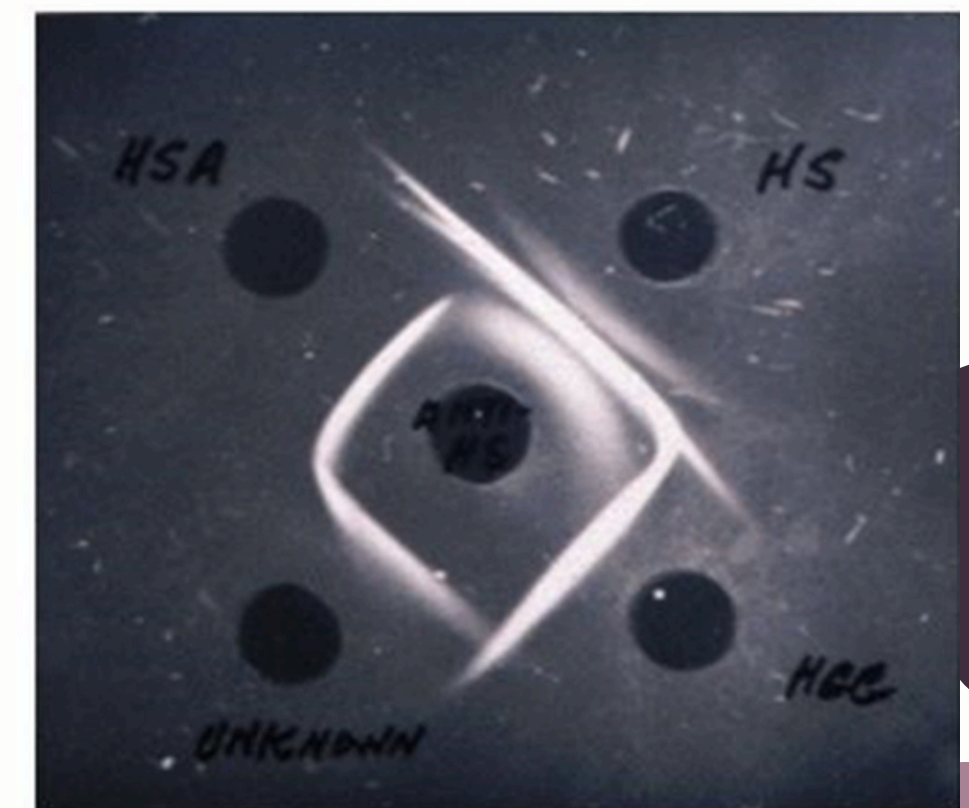
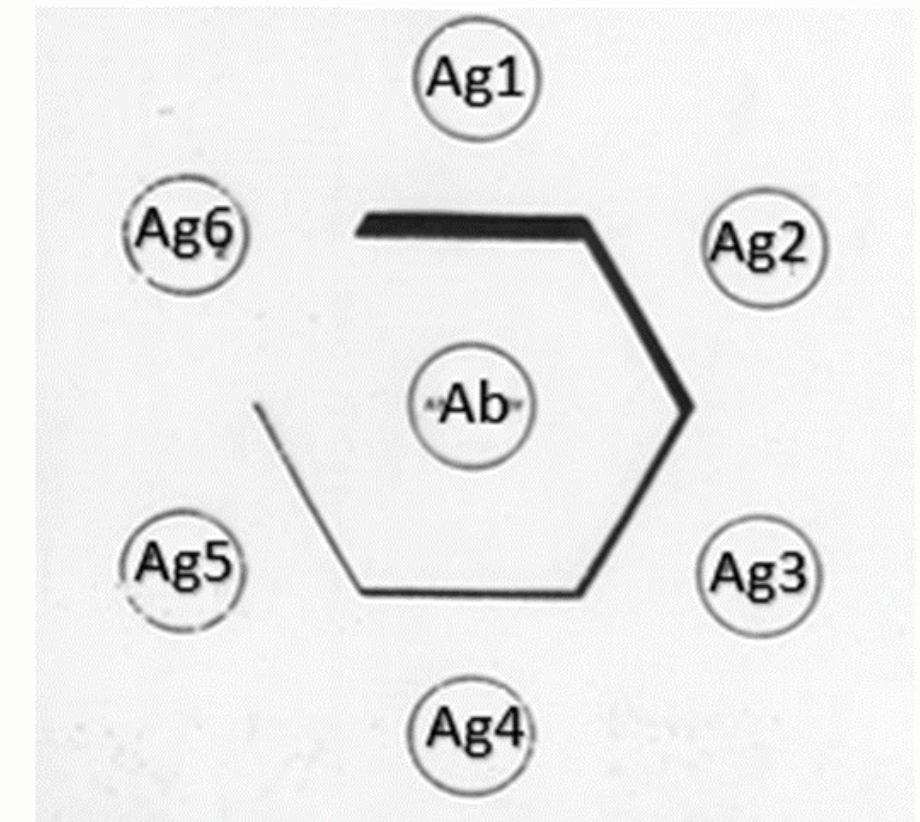
The double immunodiffusion technique could be used for semiquantitative measurement of antigen:

- 1. The thickness of the precipitation line gives an idea about the concentration**
- 2. The location of the precipitation line between the antigen and antibody also gives an idea about the concentration**

THE THICKNESS OF THE PRECIPITATION LINE

- As the concentration of antigen or antibody increases, the thickness of the precipitation line increases

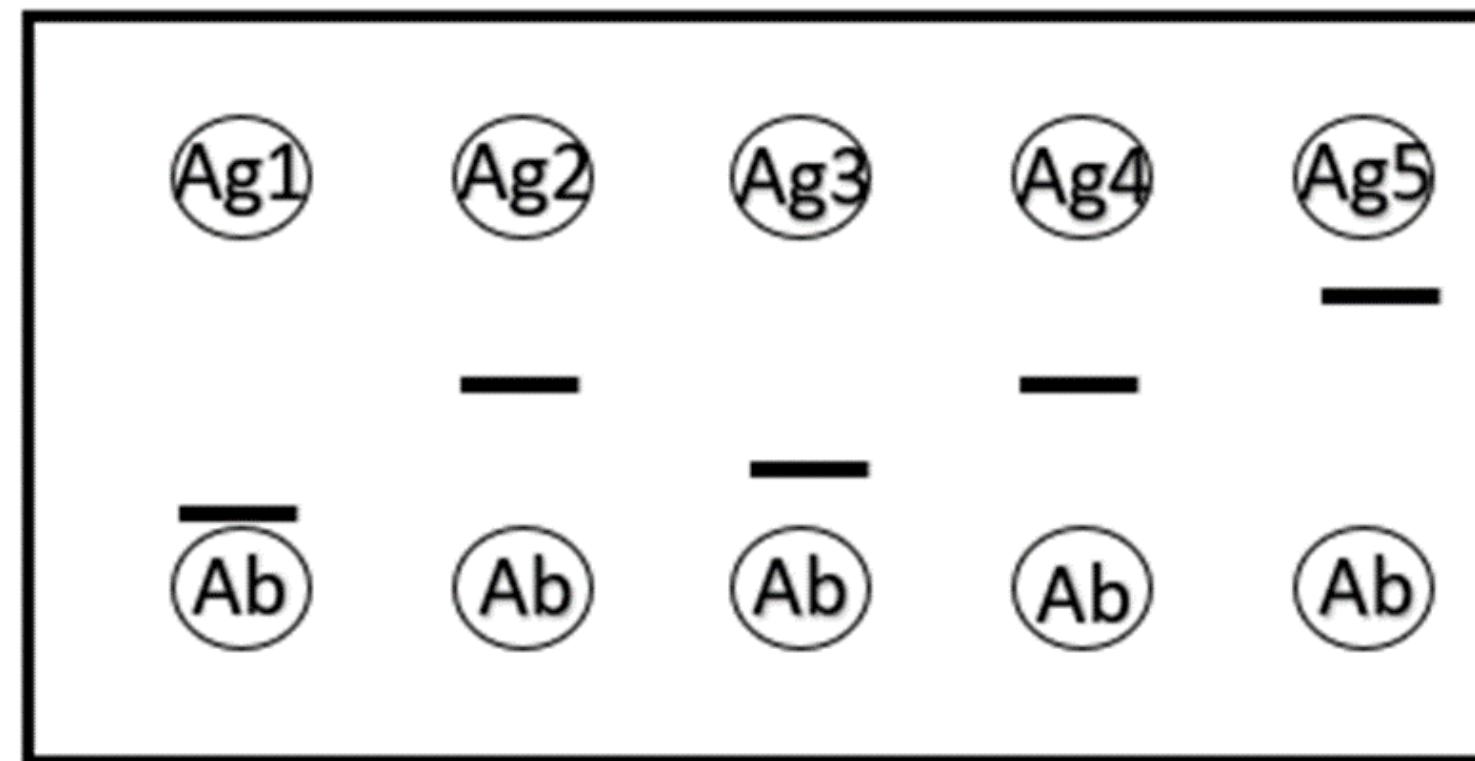
$[Ag1] > [Ag2] > [Ag3] > [Ag4] > [Ag5] > [Ag6]$



THE LOCATION OF THE PRECIPITATION LINE

- As the concentration of antigen or antibody increases, the precipitation line will move away from the well.

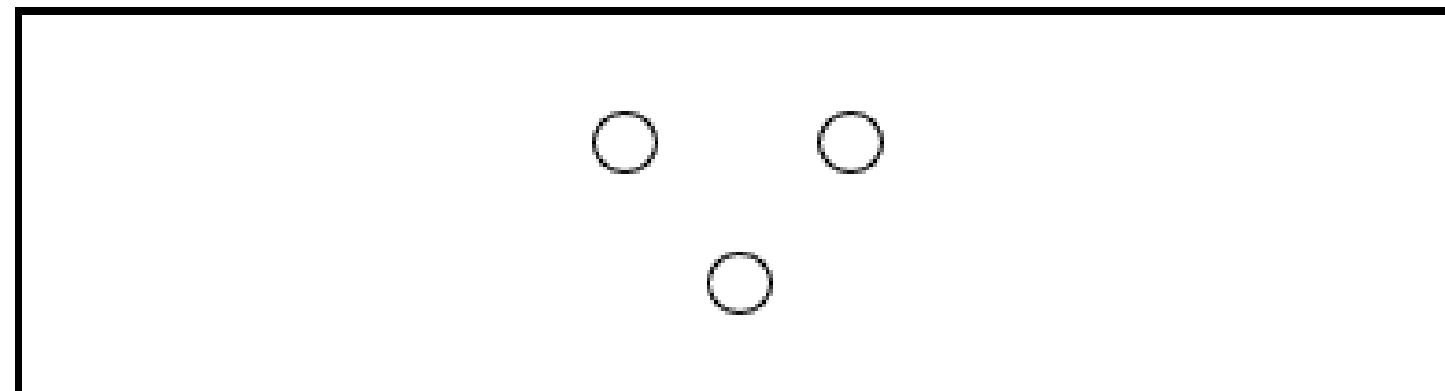
$$[Ag1] > [Ag3] > [Ag2] = [Ag4] > [Ag5]$$



Ab constant

PROCEDURE

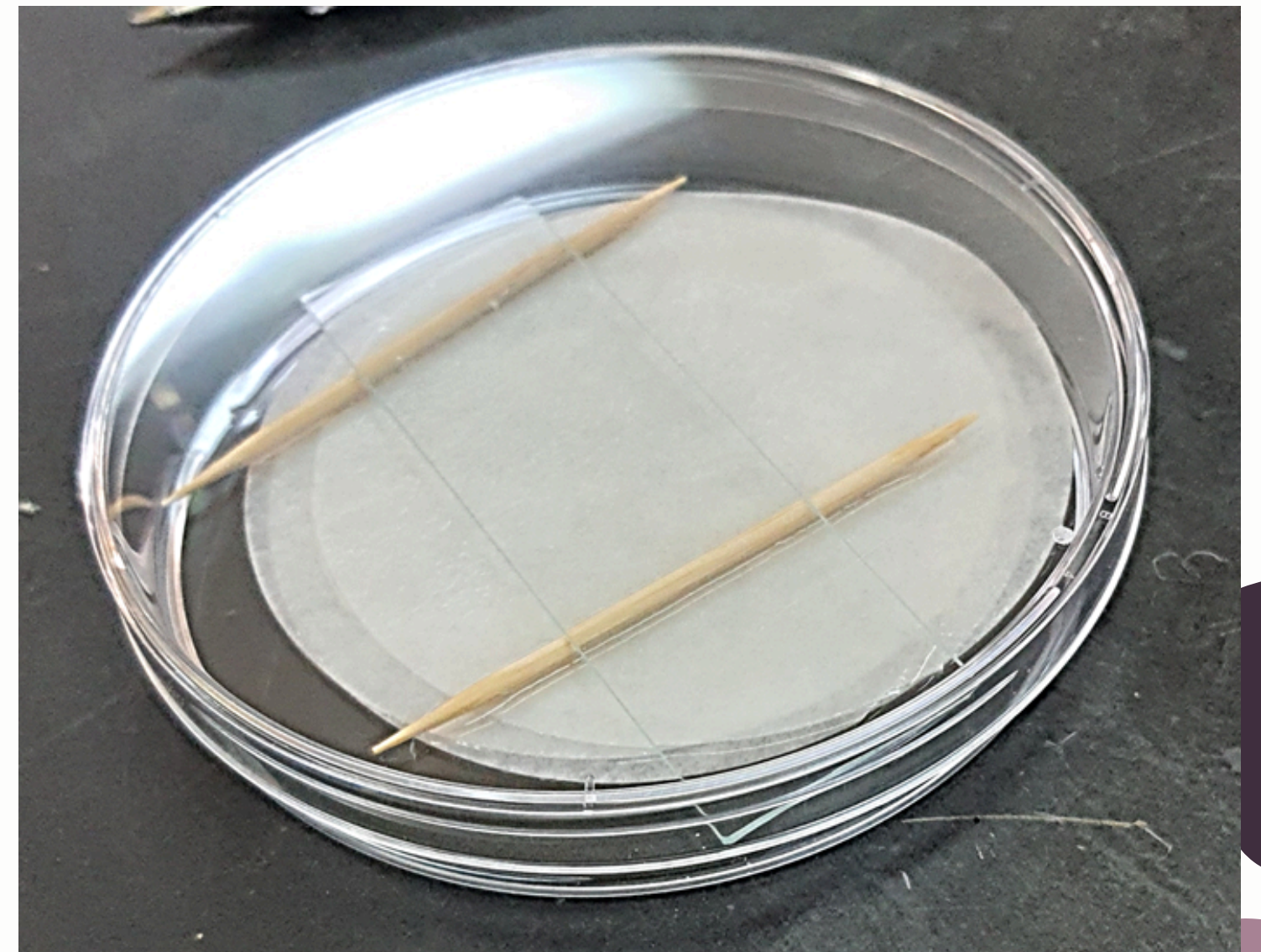
- 1. Slides are pre-coated with 1% agarose in distilled water to hold the final agarose gel in place. The slide should be completely dry. Then, a thin almost invisible plastic-like layer is formed.**
- 2. Add 5 - 6 ml of 2% melting agarose dissolved in Tris-tricine buffer to each slide.**
- 3. Leave the agarose for around 15 – 30 minutes to be cooled and solidified.**
- 4. Use an adaptor for punching holes. If the wells are not completely empty from agarose, use pipette tips or toothpicks to clean the wells.**
- 5. The antiserum (Antibody) is placed in the center well and the antigens are placed in the surrounding wells. Add 10 μl of antibody to the center well and 10 μl of antigen to each surrounding well.**

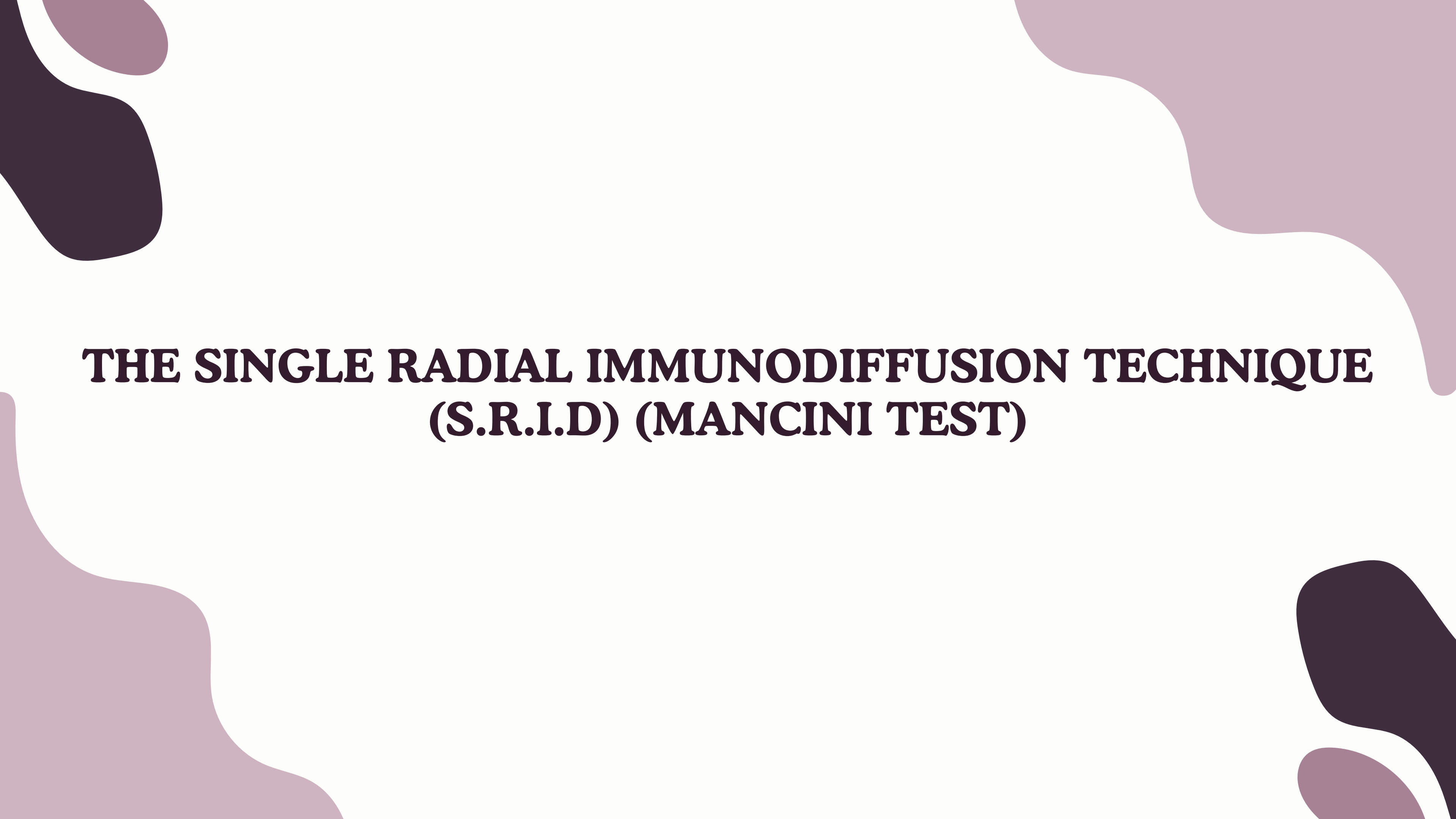


Slide containing agarose

PROCEDURE

- 6. Place the slide in a humid chamber and incubate overnight at room temperature.**
- 7. A humid chamber is prepared by covering the bottom of a Petri plate with moist paper towels and placing an unused glass slide on the moist paper towels.**
- 8. Place the immunodiffusion slide on a glass slide in the humid chamber and seal the lid of the plate with masking tape.**



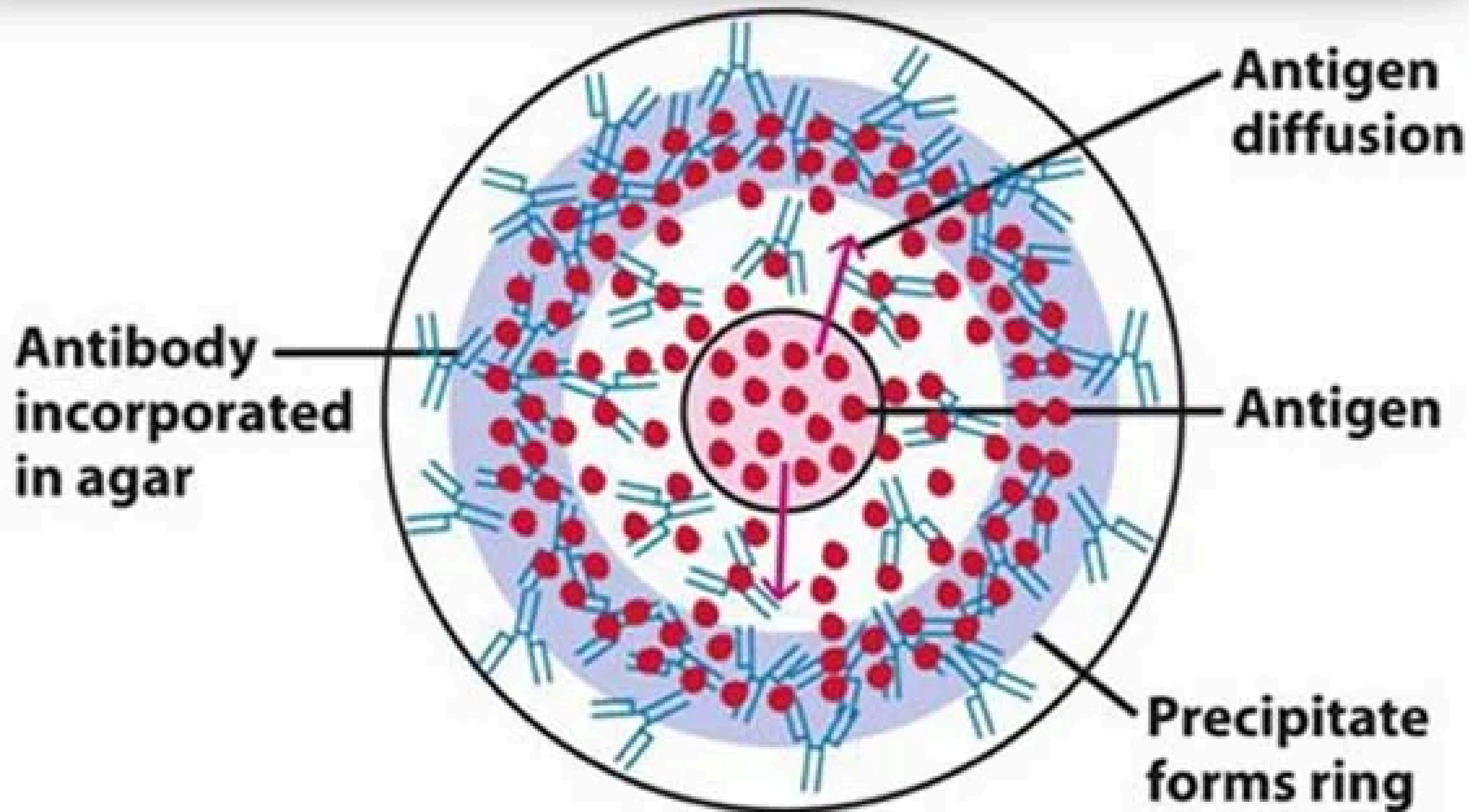


THE SINGLE RADIAL IMMUNODIFFUSION TECHNIQUE (S.R.I.D) (MANCINI TEST)

THE SINGLE RADIAL IMMUNODIFFUSION TECHNIQUE

- **The single radial immunodiffusion (SRID) technique is a quantitative method used in immunology to measure the concentration of specific antigens in a sample.**
- **It is based on the principle of antigen-antibody interaction and the formation of a visible precipitin ring.**

Radial Immunodiffusion



- Method

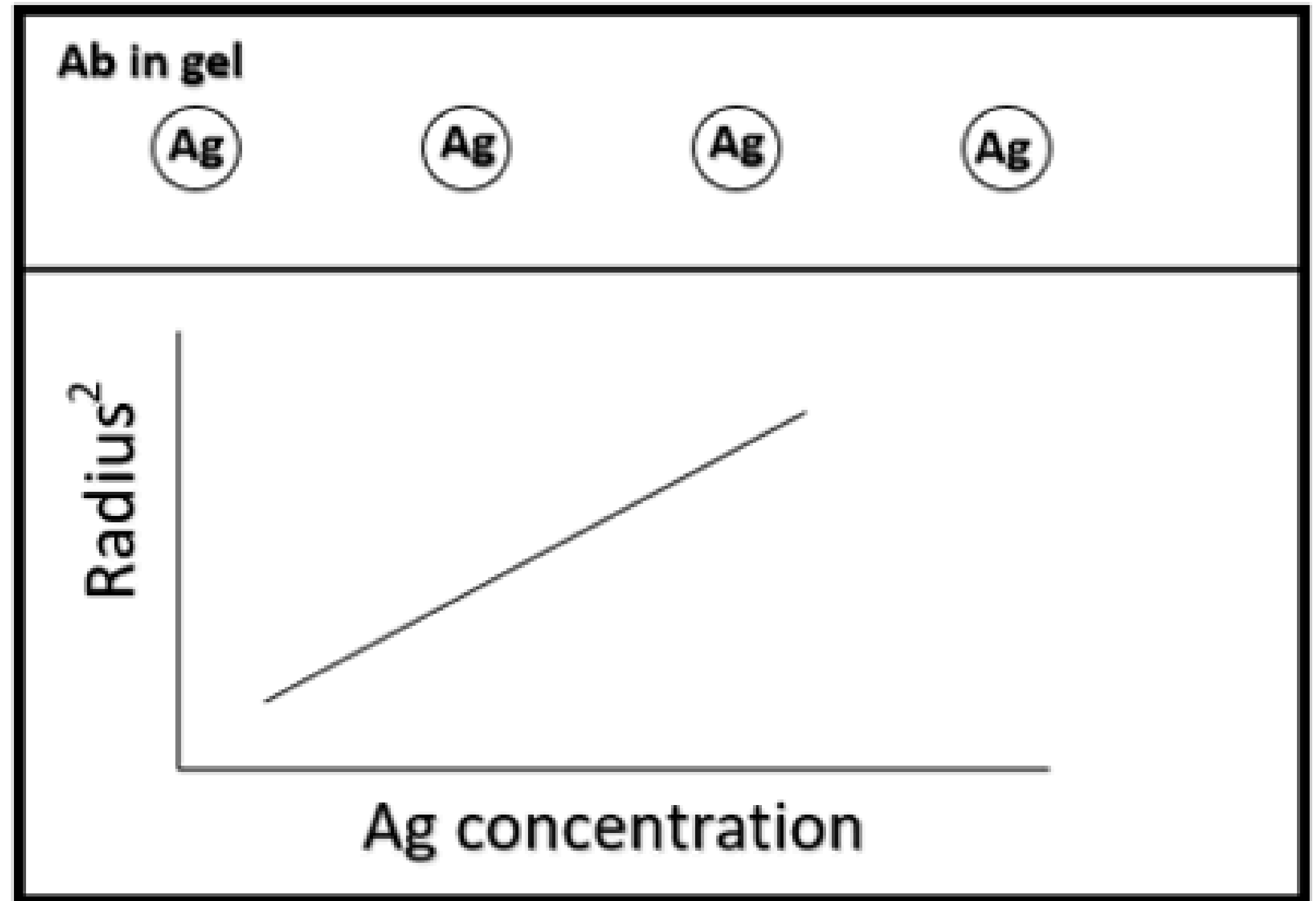
- ✓ Ab in gel
- ✓ Ag in a well

- Interpretation

- ✓ Radius of ring is proportional to the concentration

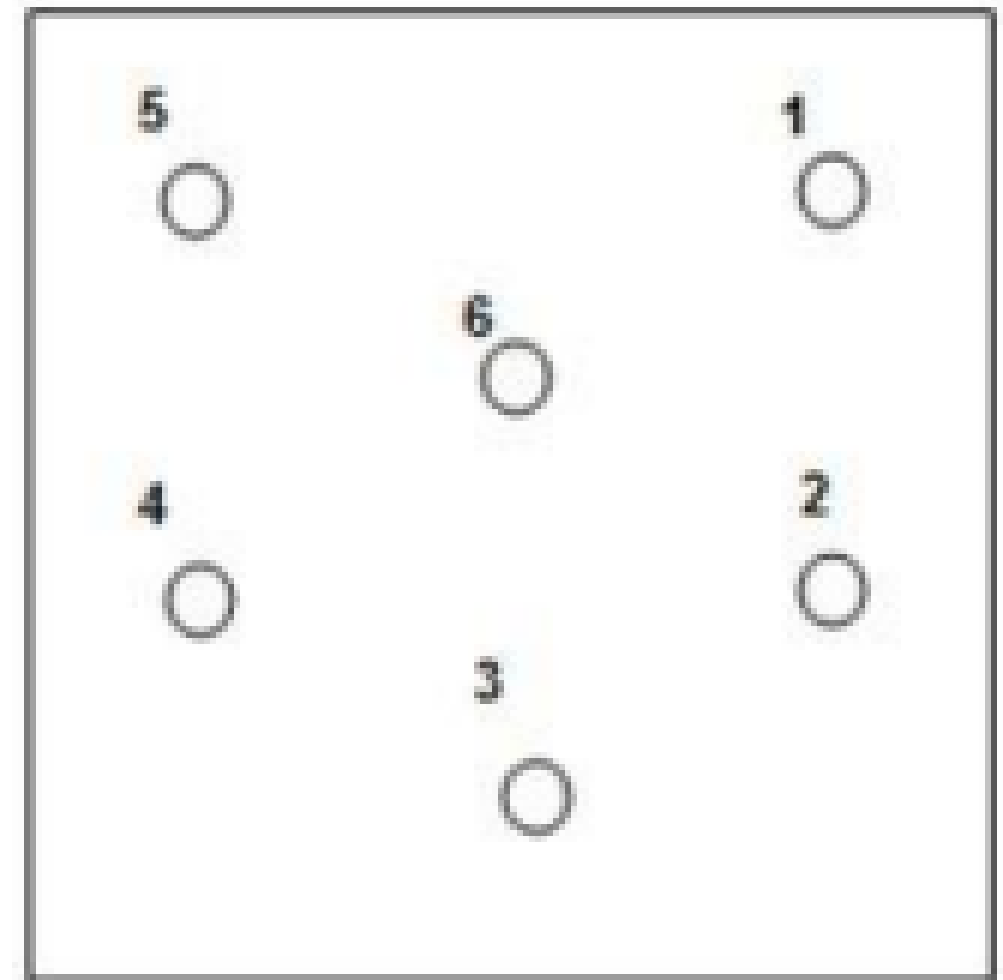
- Quantitative

- ✓ Ig levels



PROCEDURE

1. The slide must be precoated with 1% agarose.
2. 5 – 6 ml of 2% melting agarose at 52 – 53 degrees will mix with 100-120 μ l Anti-mouse IgG (Antibody against mice serum).
3. Pour the mixture over a pre-coated slide.
4. Leave the slide for 15 – 30 minutes to be cooled and solidified.
5. Cut wells using a gel puncher
6. Add 10 μ l of the given standard antigens and test antigens to the wells.
7. Keep the gel plate in a moist chamber and incubate overnight at room temperature.
8. Plot a graph of diameter of ring (on Y-axis) versus concentration of antigen (on X-axis) on a semi-log graph sheet.



PROCEDURE

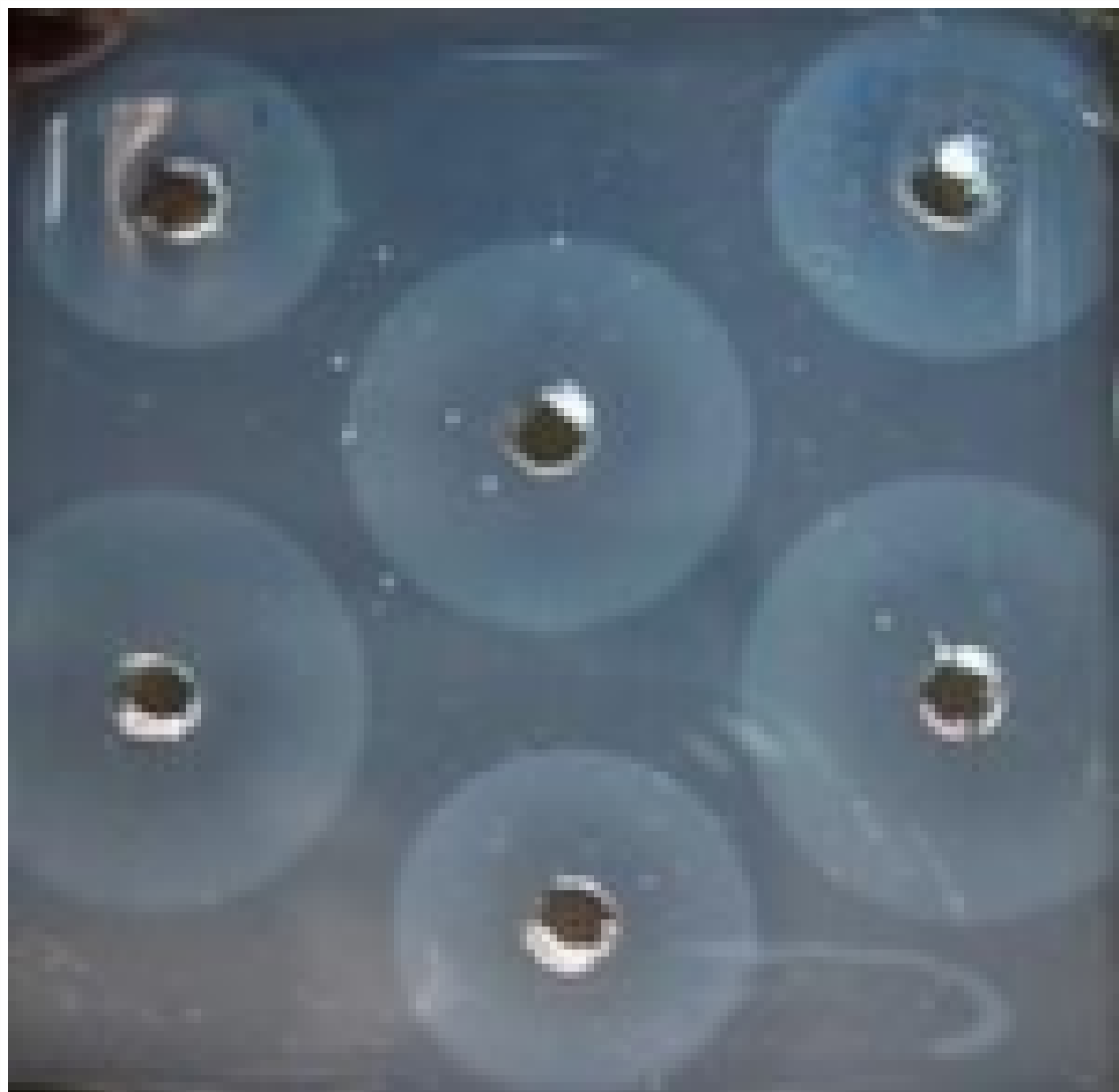
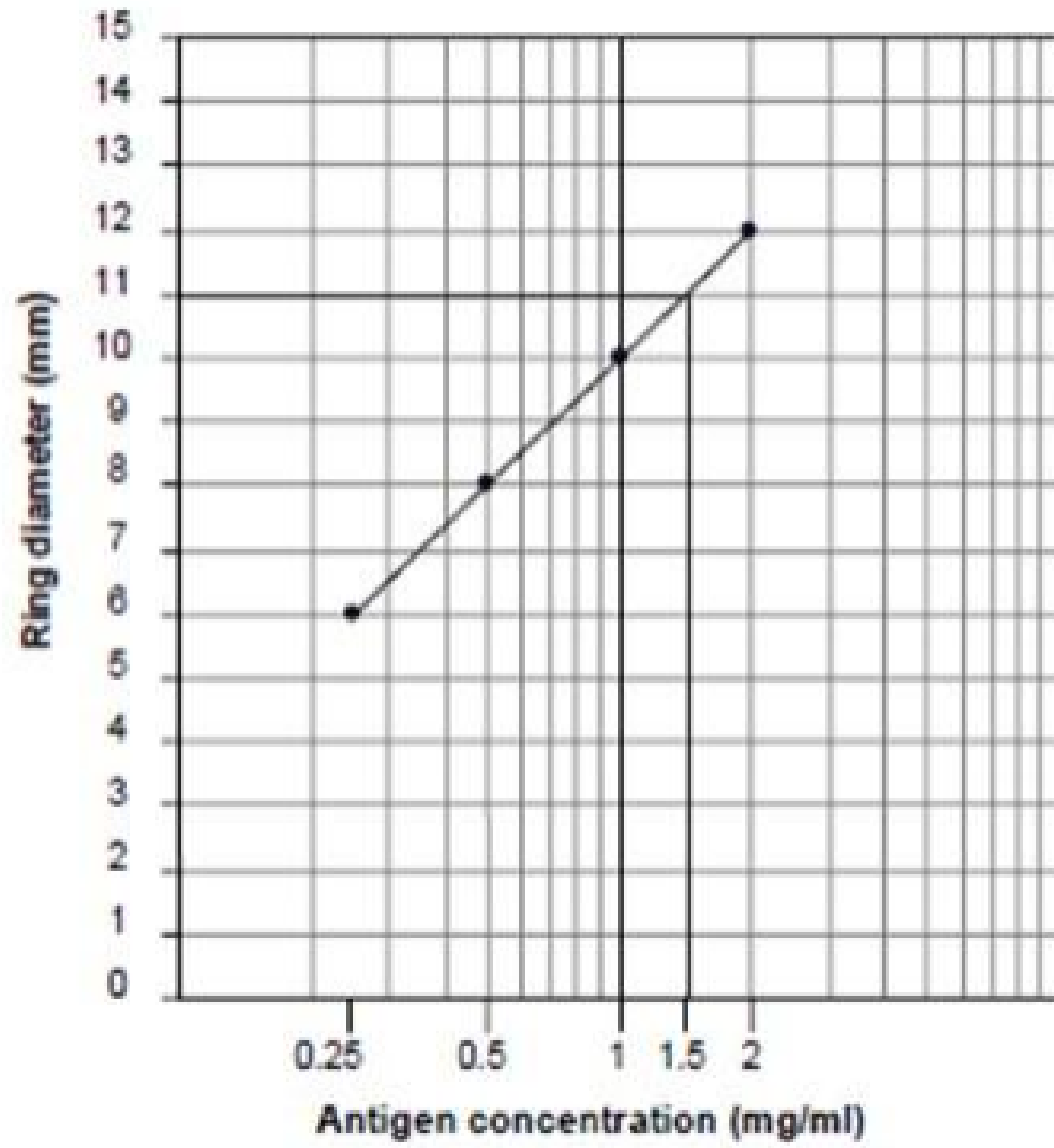
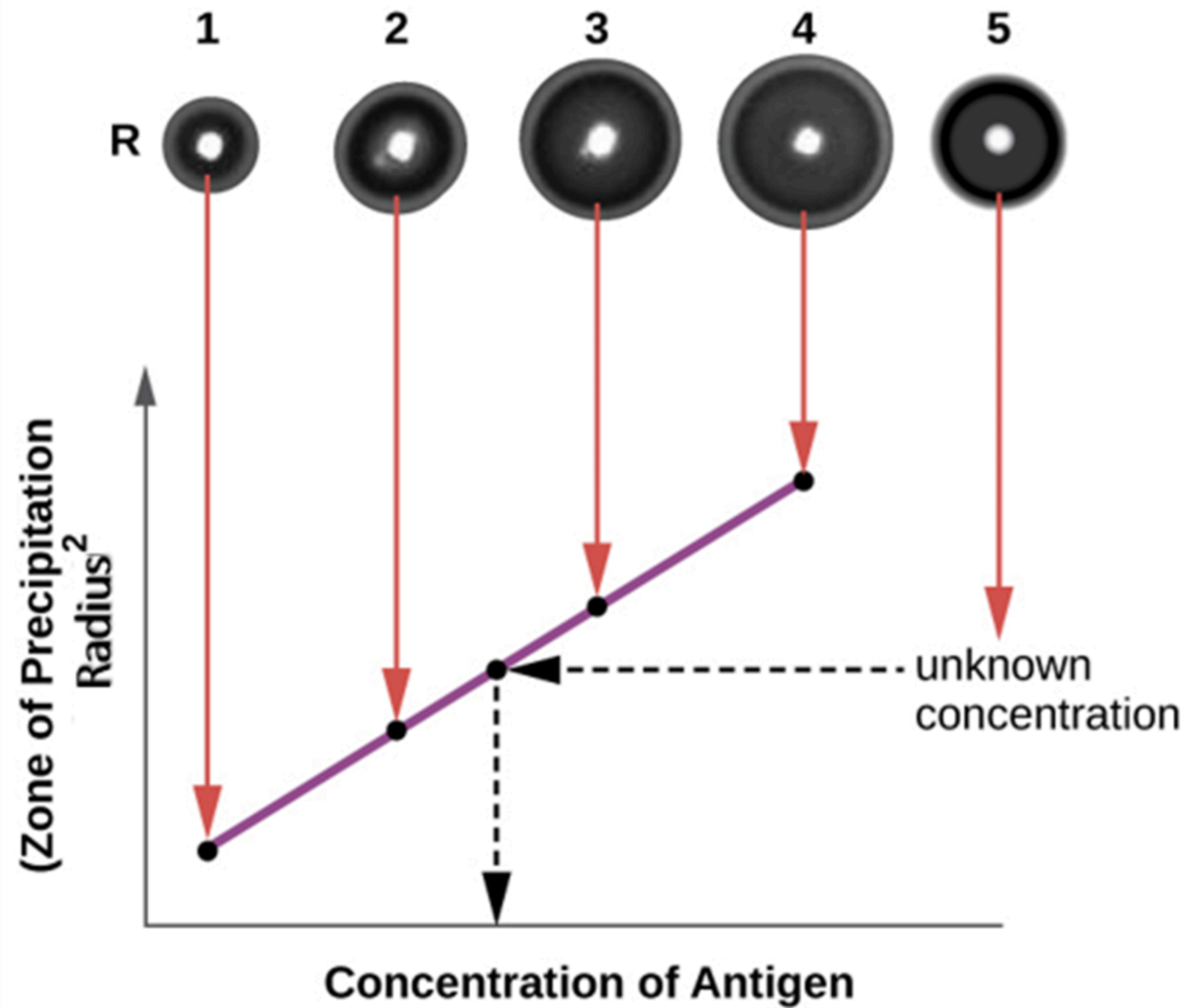


Table 1: Result of RID assay

Sample No.	Std. Ag conc. (mg/ml)	Ring diameter (mm)
A	0.25	6
B	0.5	8
C	1.0	10
D	2.0	12
Test Sample 1		11







THANK YOU